

Phase 3: Project Design Phase

System & UI Design Document

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System Architecture Flow

The system follows a standard client-server Machine Learning architecture, as illustrated below:

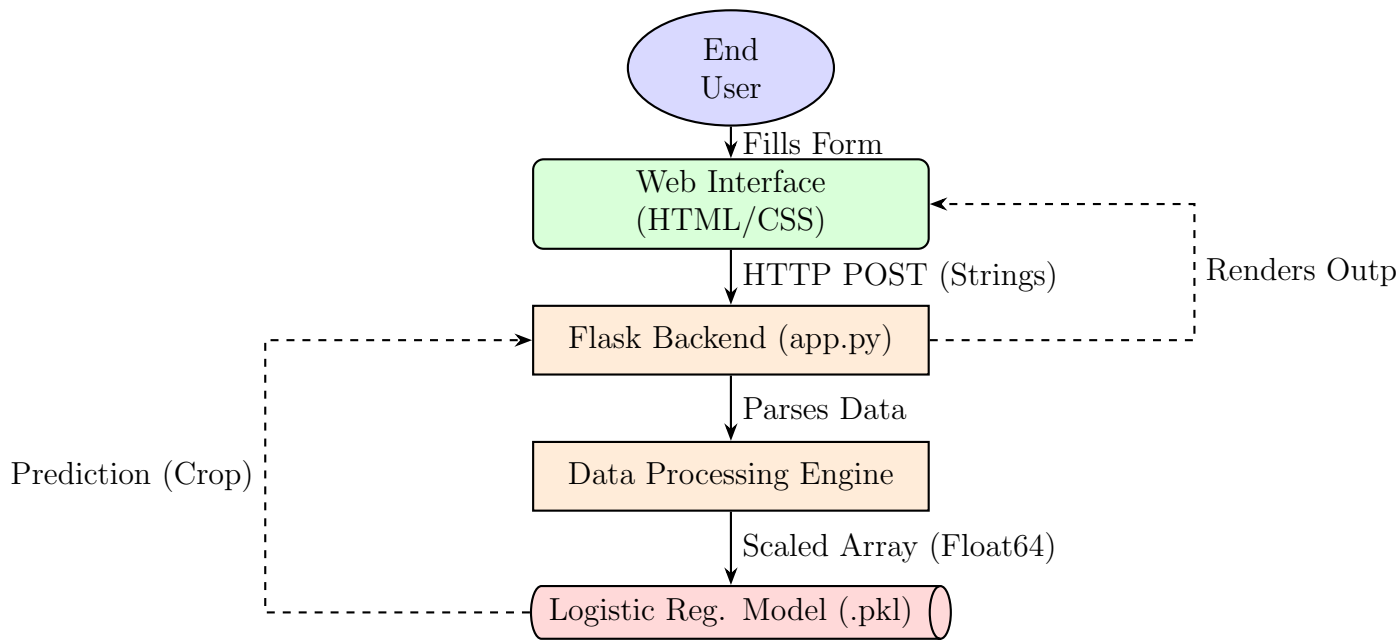


Figure 1: OptiCrop System Architecture Flow

Data Flow Logic

When user data arrives via HTTP POST, it is in string format. The Flask app executes a strict sequence of transformations to ensure the Logistic Regression model receives the exact scaled numerical distribution it was trained on, using `scaler.pkl` to prevent feature skewness.

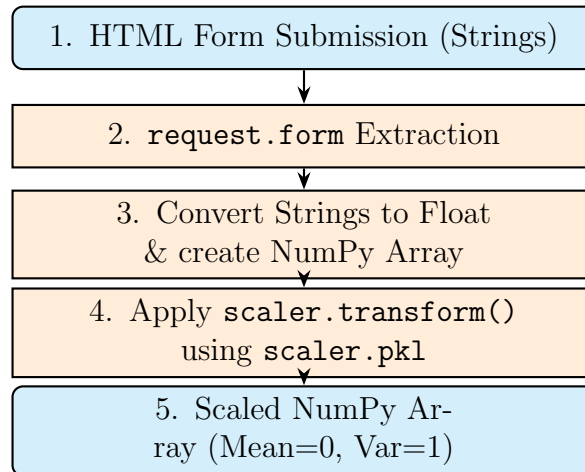


Figure 2: Data Preprocessing and Transformation Flow

Entity-Relationship (ER) Diagram

The dataset follows a single-entity structure, as all features belong to one unified agricultural observation record. The diagram below illustrates the `AGRICULTURAL_RECORD` entity and its grouped attributes.

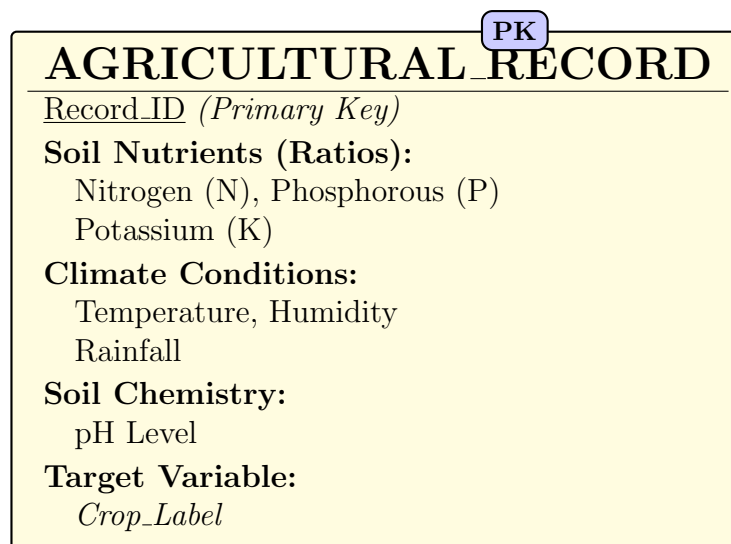


Figure 3: Entity-Relationship Diagram for Crop Dataset

User Interface Mockups

- **Home Page (index.html):** A clean landing page displaying the project title, an introductory paragraph about data-driven agriculture, and a prominent green "Get Started" button that routes to the prediction form.
- **About Page (about.html):** A text-focused layout detailing the machine learning methodology, the environmental factors utilized, and how the system benefits researchers and farmers.
- **Prediction Page (predict.html):** Features a structured HTML form containing 7 numeric input fields (N, P, K, Temperature, Humidity, pH, Rainfall) mapped to continuous variables. Below the form, a dynamic green result box appears upon successful prediction displaying the recommended crop name.